

ENCI335

Structural Analysis and Systems I

Tutorial 6 Solution

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Objectives

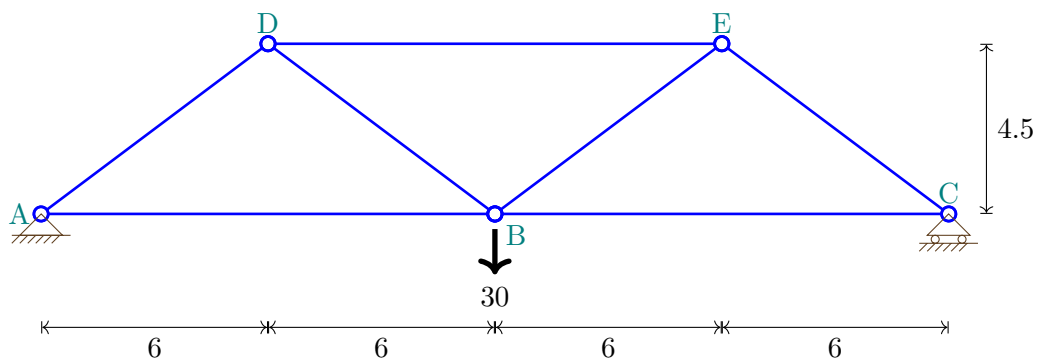
This tutorial provides an opportunity to practice the following:

- Castigliano's Theorems
- Unit Load Method

Question 1 Castigliano's Theorems

Consider the following truss, where all members have the same axial stiffness EA .

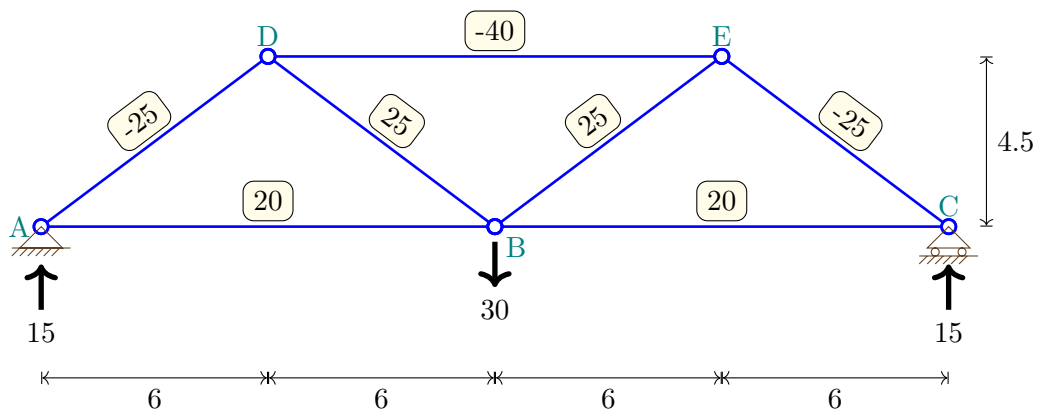
- Use the principle of conservation of energy to determine the vertical deflection at B in terms of EA .
- Use Castigliano's second theorem to determine the horizontal displacement at C in terms of EA .



Solution:

The structure and the applied load are symmetric with respect to the central axis. Therefore, the vertical reactions at A and C are both 15 (upwards). We also expect the axial forces N in all truss members to be symmetrically distributed throughout the structure. Since there is no horizontal load applied, there is no horizontal reaction at A.

We solve for the truss axial forces N using the equilibrium equations, i.e., $\sum F_x = 0$ and $\sum F_y = 0$ at all the joints. There are 5 joints in total, but due to the structure's symmetry, equilibrium equations only need to be solved for 3 of them. Once solved, the axial forces in the members resulting from the load are as follows, with compressive forces represented as negative values:



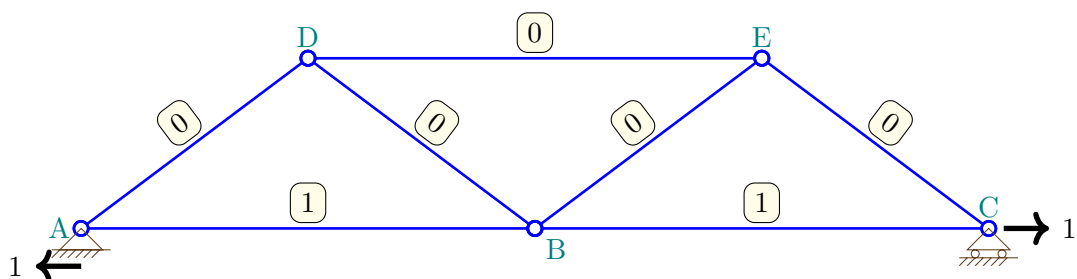
Using the energy principle:

$$W = 0.5 \cdot (-30) \cdot v_B \quad (1)$$

$$U = \sum \frac{N^2 L}{2EA} = \frac{1}{2EA} (2 \cdot 20^2 \cdot 12 + 4 \cdot 25^2 \cdot 7.5 + 40^2 \cdot 12) = \frac{23775}{EA} \quad (2)$$

$$W = U \implies v_B = \frac{-1585}{EA} \quad (\downarrow) \quad (3)$$

To find the horizontal displacement at C, we apply a virtual unit positive horizontal load at C. The resultant axial forces n in the truss members due to this load are given as:

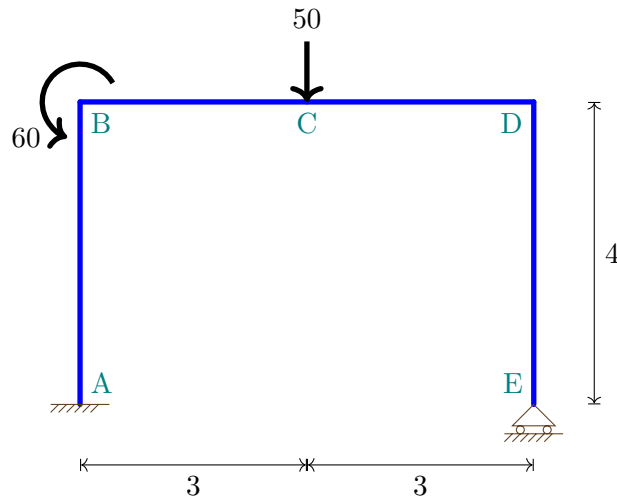


Therefore, the horizontal displacement at C is

$$u_C = \sum \frac{nNL}{EA} = \frac{1}{EA} (2 \cdot 20 \cdot 12) = \frac{480}{EA} \quad (4)$$

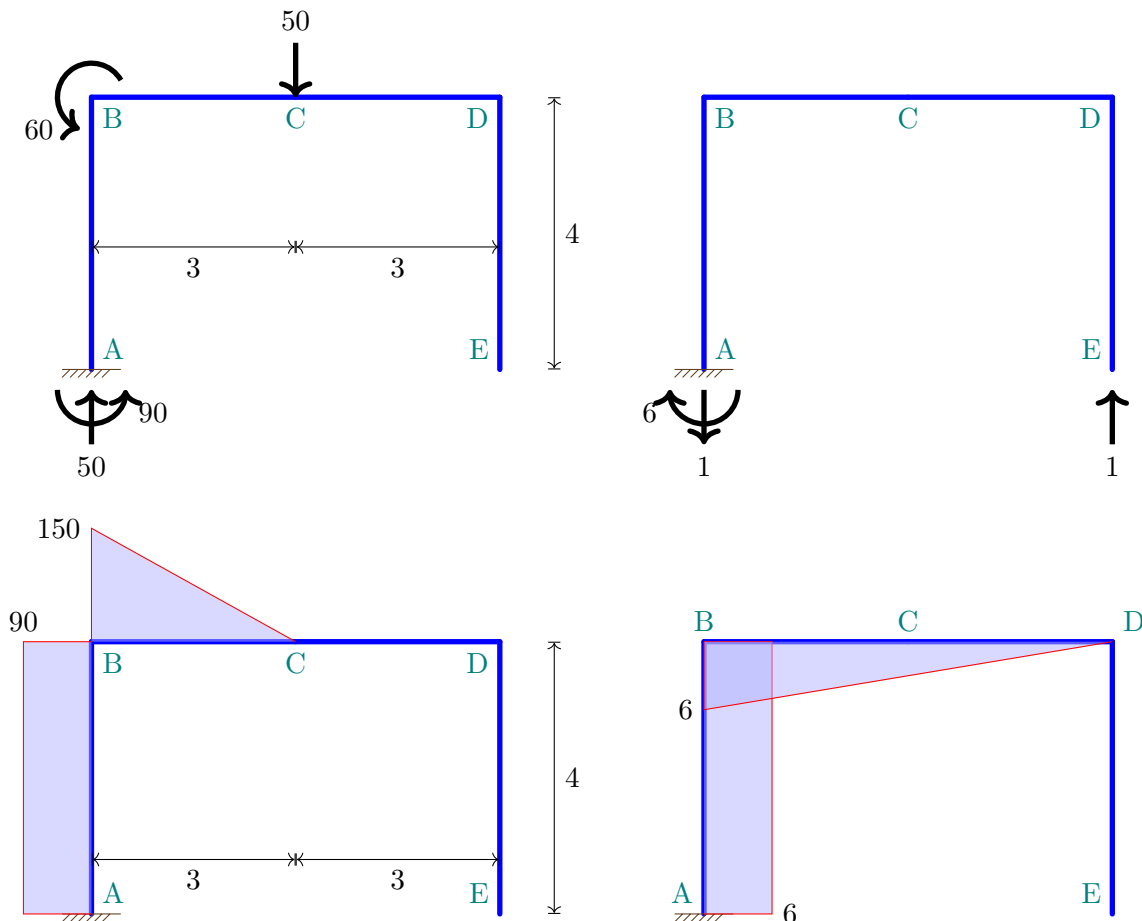
Question 2 Unit Load Method on Frames

Consider the following frame, where all members have the same flexural rigidity EI . Use the unit load method to solve for the reactions.



Solution:

This structure is the one discussed in Tutorial 4. Its SI is 1, and the vertical reaction at E is selected as the redundant force. The bending moment diagrams for the primary load case and the redundant load case are given as follows:



Using the compatibility condition that the vertical displacement at D is zero:

$$R_{Dy} \sum \int \frac{m^2}{EI} dx + \sum \int \frac{mM}{EI} dx = 0 \quad (5)$$

Since the frame has a constant flexural rigidity EI , which is a common scaling denominator in the above equation, the equation can be simplified as:

$$R_{Dy} \sum \int m^2 dx + \sum \int mM dx = 0 \quad (6)$$

Use the table given in the lectures to obtain

$$\int mM dx = -90 \times 6 \times 4 - \frac{150 \times 3}{6} (2 \times 6 + 3) = -3285 \quad (7)$$

$$\int m^2 dx = 6 \times 6 \times 4 + \frac{6 \times 6}{6} (2 \times 6 + 0) = 216 \quad (8)$$

Therefore

$$R_{Dy} = -\frac{-3285}{216} = 15.2 \quad (9)$$

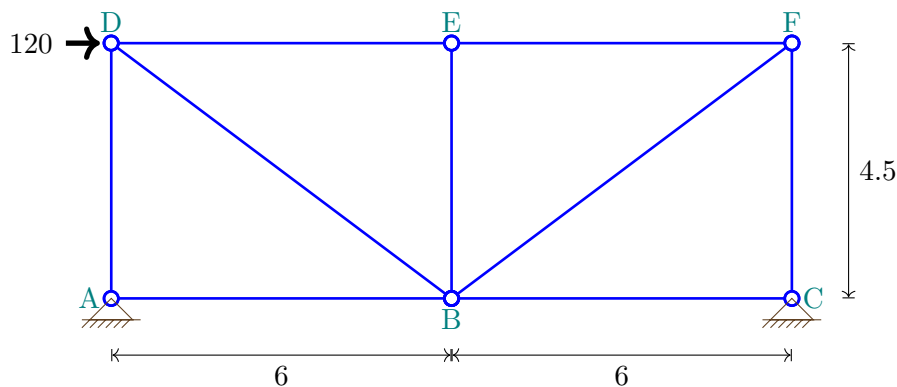
Therefore,

$$M_A = 90 - 6 \times 15.2 = -1.2, \quad R_{Ay} = 50 - 1 \times 15.2 = 34.8 \quad (10)$$

See the solution in Tutorial 4 for the final BMD, SFD, AFD, and deflected shape.

Question 3 Unit Load Method on Trusses

Consider the following truss, where all members have the same axial rigidity EA . Use the unit load method to solve for all the member axial forces.

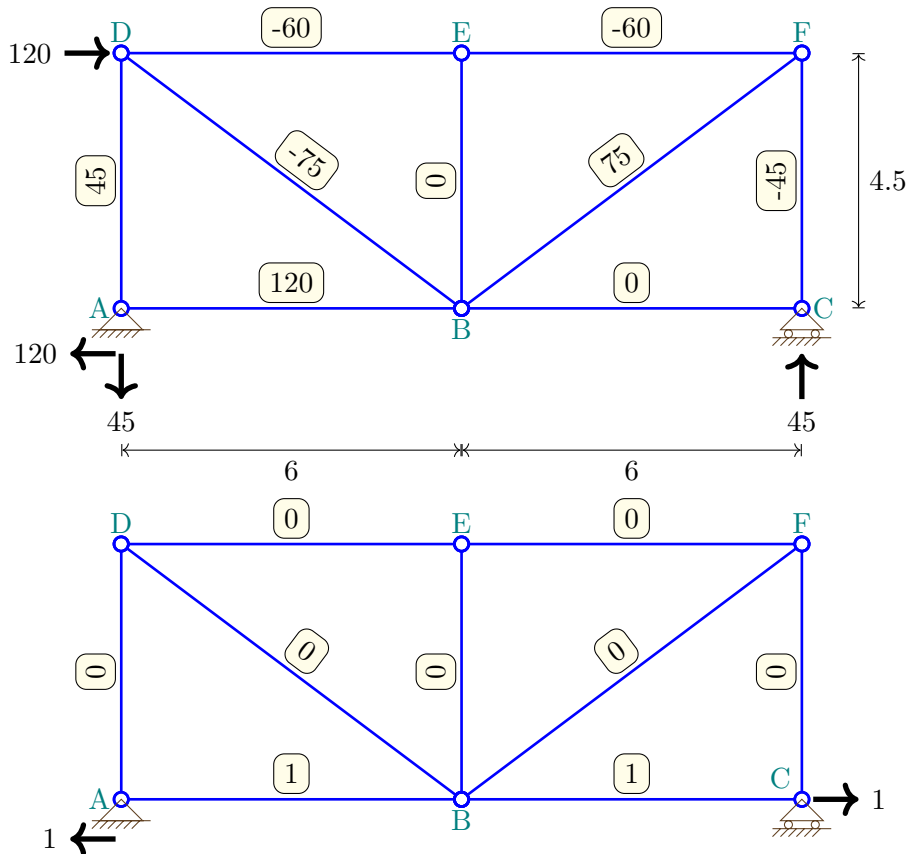


Solution:

First, calculate the SI of the truss structure:

$$n_R = 4, \quad n_M = 9, \quad n_J = 6, \quad SI = n_R + n_M - 2n_J = 1 \quad (11)$$

We choose the horizontal reaction at C to be the redundant force. We have two load cases (the primary load case and the unit load case), with their resultant reactions and member axial forces shown as follows:



Comparing the above diagrams, we see that only one member (AB) has non-zero forces in both load cases. Using the compatibility condition that the horizontal displacement at C is zero, the horizontal reaction force at C can be obtained as:

$$0 = R_{Cx} \sum \frac{n^2 L}{EA} + \sum \frac{nNL}{EA} = R_{Cx} \left(2 \times \frac{12 \cdot 6}{EA} \right) + \frac{1 \cdot 120 \cdot 6}{EA} \implies R_{Cx} = -60 \quad (12)$$

The final member axial forces in the truss structure are

